# MACHINE LEARNING 2

# 00970209

# Generative Adversarial Networks (GAN)

# Homework Assignment 4


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This report details the implementation of a Generative Adversarial Network (GAN) designed to create realistic flower images, focusing on the architecture and training methodology.

**<u>Question 1:</u>**

**Architecture**

We implemented a Deep Convolutional Generative Adversarial Network (DCGAN) architecture, following the principles introduced in the course and in Radford et al. (2016). The model consists of two networks trained adversarially: a generator (G) and a discriminator (D).

The generator receives as input a latent vector z $\sim\{N\{0,1\}$ and maps it to an image using a sequence of fully connected projection followed by transposed convolution (ConvTranspose2d) layers. These layers progressively upsample the latent representation to the target image resolution. Batch normalization and ReLU activations are used in intermediate layers to stabilize training, while a Tanh activation is applied at the output to produce normalized image values.

The discriminator is a convolutional neural network that receives either a real image from the Flowers-102 dataset or a generated image from the generator and outputs a probability indicating whether the image is real or fake. It consists of strided convolutional layers with LeakyReLU activations, gradually reducing spatial resolution while increasing feature depth, followed by a final sigmoid output.

This convolutional design leverages the strong inductive bias of CNNs for image data, allowing the model to capture local spatial structure such as edges, textures, and color patterns, which are essential for realistic flower image generation.

**Training procedure and optimization**

The model was trained using the Adam optimizer with a fixed learning rate for both the generator and discriminator. The latent space dimensionality was chosen to balance expressive power and training stability. Training was performed using mini-batches sampled uniformly from the dataset, and the same batch size was used throughout training. The choice of these hyperparameters was guided by standard practices for DCGAN training and validated empirically through monitoring loss curves and sample quality.

The GAN was trained using the standard minimax adversarial objective, where the discriminator is optimized to distinguish real from generated images, while the generator is optimized to fool the discriminator. Training alternates between updating the discriminator and the generator, following the two-player game formulation discussed in the course.

Both networks were optimized using the Adam optimizer, which combines momentum and adaptive learning rates. Adam was chosen due to its empirical stability in GAN training and its ability to handle noisy and highly non-stationary gradients, as discussed in the optimization lectures. A single learning rate was used for

both networks, and training was performed using mini-batches sampled from the dataset.

The training process was monitored by tracking the generator and discriminator losses per epoch, as well as by visually inspecting generated samples at regular intervals. This qualitative monitoring is crucial in GANs, since loss values alone do not always correlate with perceptual image quality.

**Design choices and challenges**

One of the main challenges in training GANs is maintaining a balance between the generator and discriminator. If the discriminator becomes too strong, the generator receives vanishing gradients; if the generator dominates, the discriminator fails to provide meaningful feedback. The chosen architecture depth and optimization settings were selected to keep both networks learning at a comparable pace.

Another challenge is the high diversity and complexity of the Flowers-102 dataset, which contains 102 visually distinct flower categories with large intra-class variation. This makes the generation task significantly harder than single-class datasets and increases the risk of unstable training or poor image quality in early stages.

**Sensitivity to hyperparameters**

The model was found to be moderately sensitive to hyperparameter choices, particularly the learning rate and the relative training dynamics of the generator and discriminator. Small changes in the learning rate affected convergence speed and image sharpness. However, once a stable configuration was reached, training proceeded smoothly and produced consistent improvements in image quality over time.

Batch size also influenced training stability: smaller batches introduced more noise into gradient estimates, sometimes slowing convergence, while larger batches improved stability but increased computational cost.

**Training instabilities**

Common GAN instabilities, such as mode collapse, were monitored throughout training. Based on the diversity of generated samples and the absence of repeated identical outputs, no severe mode collapse was observed. The generator produced images with varied colors, shapes, and textures across training epochs, indicating that multiple modes of the data distribution were captured.

While some artifacts and blurring remained in generated images, these are typical for GANs trained on complex, multi-class datasets and do not indicate catastrophic instability. Overall, the loss curves and qualitative results suggest that adversarial training remained stable and effective throughout the training process.

**<u>Question 2:</u> Discussion of the training loss plots**

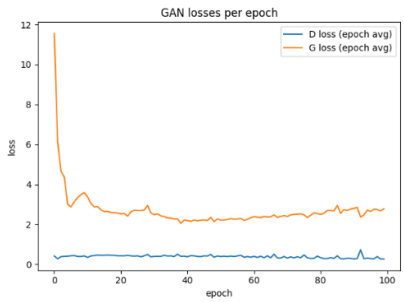
Figure 1 presents the discriminator and generator losses as a function of the training epochs. The loss curves exhibit the characteristic behavior of adversarial training in GANs and provide insight into the convergence properties of the model.

At the beginning of training, the generator loss is very high, while the discriminator loss is relatively low. This behavior is expected, as the generator initially produces samples that are far from the data distribution, allowing the discriminator to easily distinguish between real and fake images. Consequently, the discriminator assigns low probabilities to generated samples, leading to a large generator loss.

As training progresses, the generator loss decreases sharply, indicating that the generator rapidly improves its ability to produce more realistic images. During the same period, the discriminator loss remains low and relatively stable, suggesting that the discriminator continues to provide meaningful gradients without becoming saturated. This phase corresponds to the generator learning coarse visual structures and basic color distributions, which is also reflected in the visual evolution of generated samples.

After the initial transient phase, both losses enter a stable regime. The generator loss fluctuates mildly around a moderate value, while the discriminator loss stays consistently low with small oscillations. These oscillations are typical in GAN training and reflect the ongoing adversarial game between the two networks rather than divergence or instability. Importantly, there is no evidence of exploding losses or collapse of either network.

Figure1:




**Convergence analysis**

Although GANs do not converge in the traditional sense of minimizing a single objective, the observed loss behavior indicates practical convergence. The stabilization of both generator and discriminator losses, together with the continuous improvement and eventual stabilization of generated image quality, suggests that the model reached a dynamic equilibrium where neither network dominates the other.

The absence of strong oscillations, divergence, or sudden loss spikes indicates that the adversarial training process remained stable throughout training. Combined with the qualitative results, the loss curves support the conclusion that the training converged to a meaningful solution in which the generator captures multiple modes of the data distribution.

**Question 3:**

**Conclusions**

In this work, we implemented and trained a convolutional GAN on the Flowers-102 dataset with the goal of generating diverse and visually plausible flower images. The conclusions presented here are based directly on the observed training behavior, generated samples, and latent space analyses obtained during the experiments.

First, the training process proved to be stable and effective. The discriminator and generator loss curves exhibited the expected adversarial dynamics, with an initial transient phase followed by stabilization. While GAN training does not converge to a single minimum, the observed loss behavior, together with the qualitative improvement of generated images over time, indicates that the model reached a meaningful equilibrium. No signs of divergence or severe instability were observed.

Second, the quality and diversity of generated images improved consistently throughout training. Early samples consisted of noisy, unstructured patterns, while later samples displayed clear flower-like shapes, coherent color distributions, and varied textures. Importantly, the generated images did not collapse to a single appearance or color scheme. The diversity observed across samples suggests that the generator successfully captured multiple modes of the data distribution, despite the complexity of the multi-class Flowers-102 dataset.

Third, the latent space analysis provides additional evidence that the model learned a structured representation. Visually similar image pairs generally corresponded to smaller image-space distances and tended to be closer in latent space than dissimilar pairs. Although the separation was not perfect, the PCA visualization of latent vectors showed that similar samples cluster more closely than dissimilar ones. This indicates that proximity in the latent space partially reflects perceptual similarity, even though the mapping from latent space to image space is highly non-linear.

At the same time, several limitations were observed. Some generated images still exhibit blur, artifacts, or unrealistic details, particularly in fine-grained structures. In addition, the overlap between latent representations of different samples suggests that the latent space is not fully disentangled. These limitations are expected given the complexity of the dataset and the inherent challenges of GAN training.

Overall, the results demonstrate that the implemented GAN successfully learns to generate diverse flower images and to organize the latent space in a semantically meaningful way. While there is room for improvement in image sharpness and latent disentanglement, the model meets the objectives of the assignment and provides a solid demonstration of adversarial generative learning in practice.

**<u>Question 4:</u>**

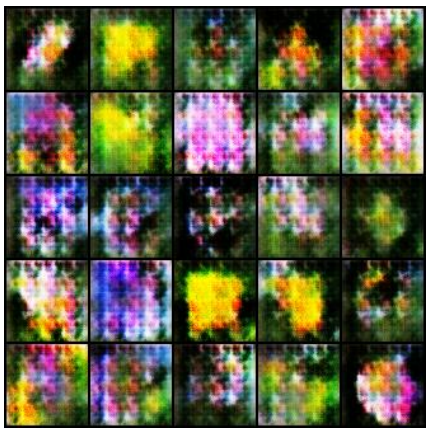
**Generated images and qualitative evaluation**

**4.1 Early training stage – initial samples**

Figure 1 shows generated images from the early stages of training. At this stage, the samples appear highly noisy and lack recognizable structure. Colors are present, but there is no clear separation between foreground and background, and no flower-like shapes can be identified. This behavior is expected, as the generator has not yet learned meaningful features from the data distribution.

Quality: Very low – dominated by noise, no semantic structure.

Figure 1**:**



**4.2 Intermediate training stage – emergence of structure**

Figure 2 presents samples generated at an intermediate stage of training. At this point, coarse structures begin to emerge. Some images display blob-like shapes and rough color regions that resemble flower components. However, the images still suffer from blurriness and artifacts, and fine-grained details are largely missing.

Quality: Moderate – the generator begins to capture coarse spatial structure and dominant color regions, but images remain blurry and lack fine-grained details. Foreground–background separation is still weak, and artifacts are common.

Figure 2**:**

### 4.3 Final training stage – image quality and diversity

Figure 3 shows samples generated at the final stage of training. The images exhibit clear flower-like structures, coherent color distributions, and improved foreground–background separation. Many samples capture characteristic visual patterns of flowers, such as petal arrangements and dominant color regions.

Despite these improvements, some limitations remain. Certain images still contain blurring, local artifacts, or unrealistic textures, particularly in fine details. These imperfections are common in GANs trained on complex, multi-class datasets such as Flowers-102.

Quality: Good but not perfect – generated images are clearly recognizable as flowers, with coherent color distributions and improved structural consistency. Many samples exhibit petal-like shapes and meaningful composition. However, fine details such as sharp edges and textures are often missing, and some images still contain local artifacts or blur, especially in complex regions.

Figure 3: final sample

### 4.4 Diversity and overall assessment

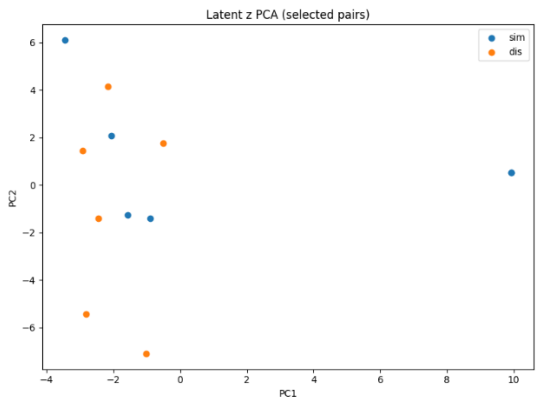
Across the final samples, a high level of diversity is observed. The generated images vary significantly in color, shape, and overall appearance, suggesting that the model does not suffer from severe mode collapse. The generator is able to produce a wide range of flower types rather than repeatedly generating a small set of similar images.

Overall, the qualitative results indicate that the GAN successfully learned a meaningful generative model of the data. While image sharpness and fine details could be further improved, the model captures essential visual characteristics of the Flowers-102 dataset.

## Question 5:

**Latent space visualization.**
Figure X shows a two-dimensional visualization of the latent vectors ( z ) corresponding to selected pairs of generated images, obtained using PCA for dimensionality reduction. The visualization includes pairs of visually similar images and pairs of visually dissimilar images, marked with different colors.

**Figure 1:** PCA projection of latent vectors corresponding to visually similar (blue) and dissimilar (orange) image pairs.




**Analysis based on the plot.**
From the PCA projection, latent vectors corresponding to visually similar image pairs tend to appear closer to each other in the projected space, while vectors corresponding to dissimilar image pairs are generally more separated. Although some overlap exists between the two groups, the overall trend suggests that the latent space captures meaningful information related to visual similarity.

**Analysis based on L2 distances.**
This observation is further supported by quantitative measurements using the ( L2 ) norm in the original latent space. Visually similar image pairs consistently exhibit smaller image-space distances and, in most cases, smaller latent-space ( L2 ) distances compared to dissimilar pairs. In contrast, dissimilar image pairs show significantly larger image-space distances and generally larger latent-space distances.

While the separation in latent space is not perfect, the combined qualitative and quantitative results indicate that proximity in the latent space partially reflects perceptual similarity between generated images.

**Figure 2 — Similar image pair**

**Dissimilar image pair**

**Similar image pair**

**Conclusion.**
Overall, the latent space learned by the GAN is structured rather than random. Similar images tend to be mapped to nearby regions of the latent space more often than dissimilar images, although the relationship is not strictly linear. This behavior is expected in GANs, where the mapping from latent vectors to images is highly non-linear and not explicitly constrained to enforce semantic clustering. Therefore, based on both the PCA visualization and the measured (L_2) distances, latent vectors of similar images are closer to each other than those of dissimilar images in most cases.